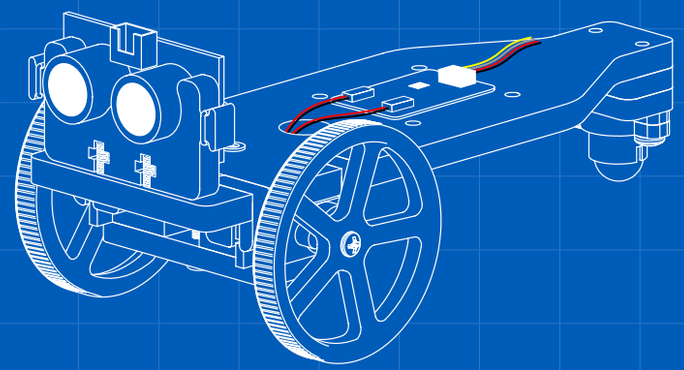
PIPER

Mission:

Piper Make Robotic Rover

SIDE A
Assembly

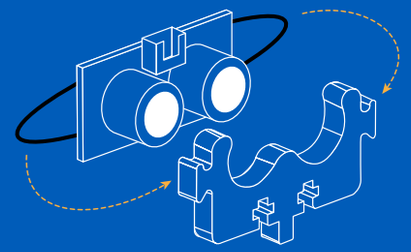


INVENTORY

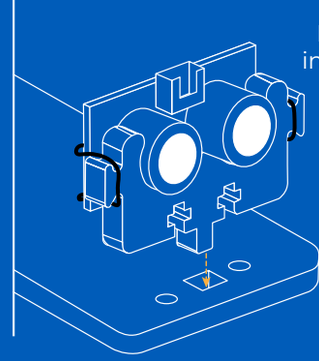
- 1 Base
- 2 Motor Spacers
- 1 Motor Holder
- 2 #1 Spacers
- 2 #2 Spacers
- 2 FM90 Geared Motors
- 1 Motor Module
- 2 Wheels and Rims
- 2 4-Pin Cables
- 1 Range Finder Holder
- 1 Ball and Holder
- 2 #4-40 Screws (A)
- 2 #2-56 Screws (B)
- 2 #4-40 Screws (C)
- 2 Small Screws (D)
- 2 #4-40 Nuts
- 1 O-Ring
- 1 Velcro Pieces

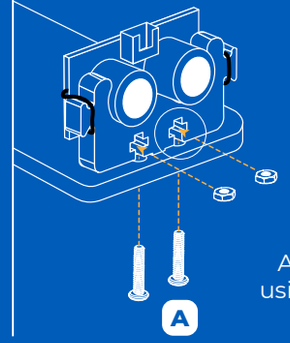
1

Use O-Ring to attach Range Finder to Holder



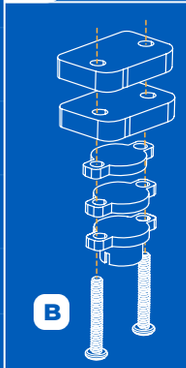
Insert Holder into Rover Base





Attach Holder using #4-40 Nuts

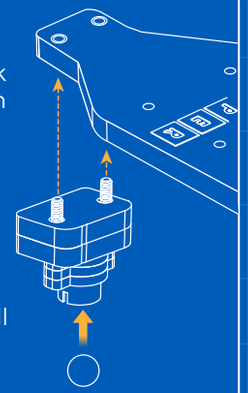
2



Remove Ball from Holder

Insert Screws as shown

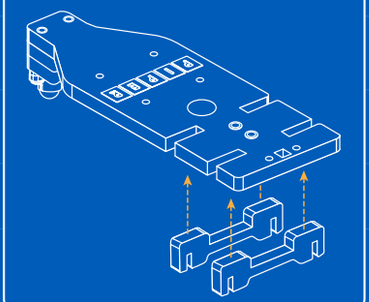
Attach stack to Base with Screws



Reinsert Ball into Holder

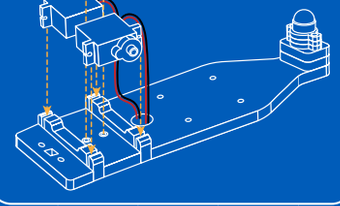
3

Insert Motor Spacers into Base notches



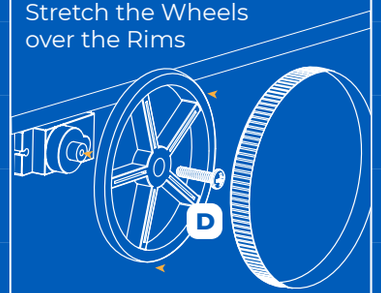
4

Orient Motors so wires extend through the hole as shown



5

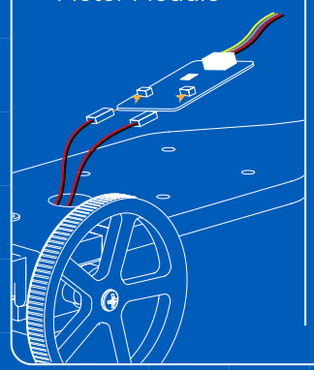
Stretch the Wheels over the Rims



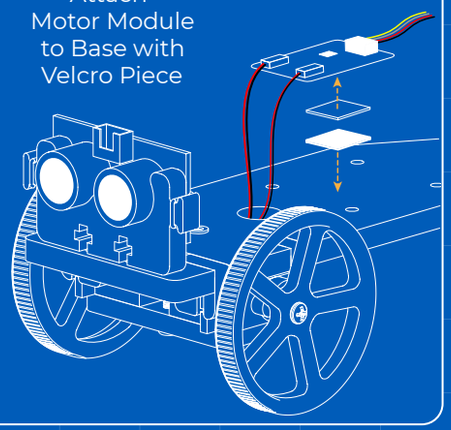
Insert the Motor shaft into the ridged hole of the Wheel

6

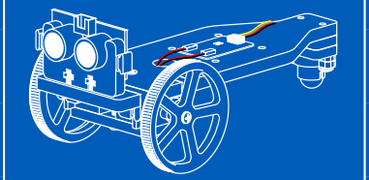
Connect wires to Motor Module



Attach Motor Module to Base with Velcro Piece



ASSEMBLY COMPLETE!



Head to Piper Make to continue your mission

make.playpiper.com

Piper Make Robotic Rover - Version 2.0.0

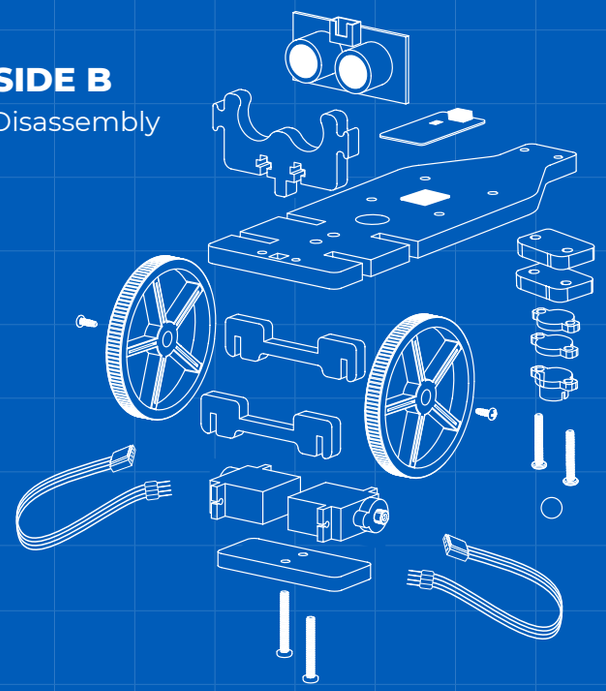
PIPER

Mission:

Piper Make Robotic Rover

SIDE B

Disassembly



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- 1 Ball and Holder
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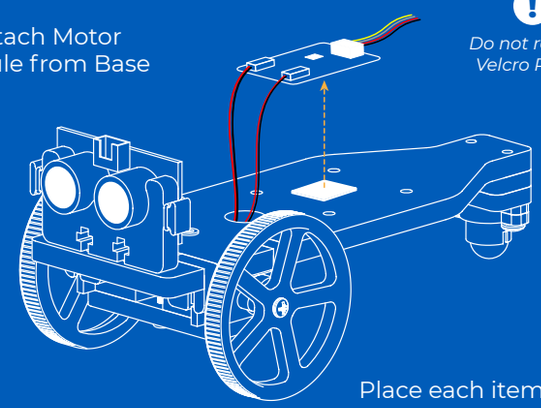
- 2 #4-40 Screws (C)
- 2 Small Screws (D)
- 2 #4-40 Nuts
- 1 O-Ring
- 1 Velcro Pieces

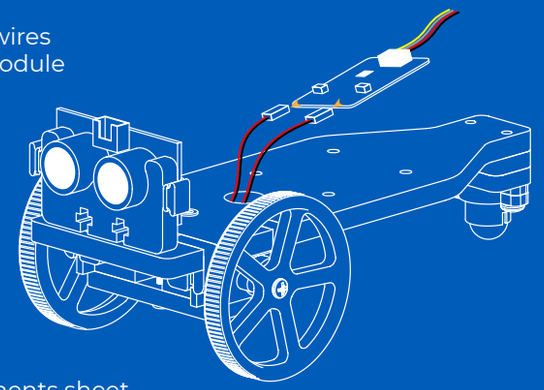
1

Detach Motor Module from Base



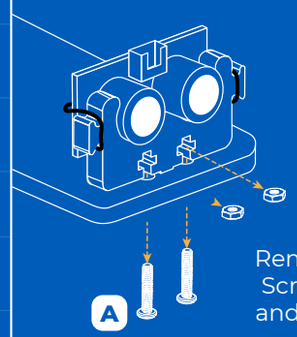
! *Do not remove Velcro Pieces*

Disconnect wires from Motor Module



Place each item on its spot on the Components sheet

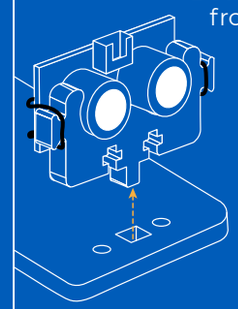
2



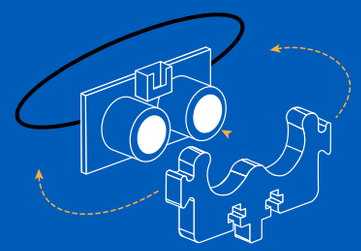
A

Remove Screws and Nut

Detach Holder from Rover Base



Remove O-Ring to detach Range Finder from Holder



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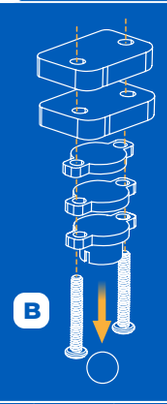


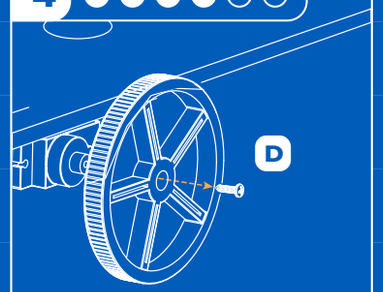
B

Remove Ball from Holder

Remove Screws

Reinsert Ball into Holder for storage

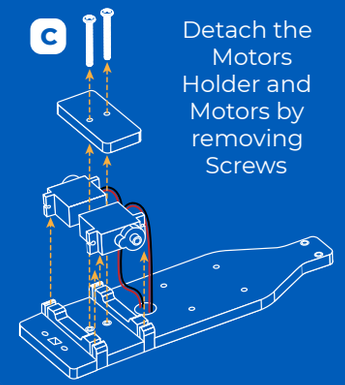
4



D

Detach Wheels from the Motors by removing Screws

5

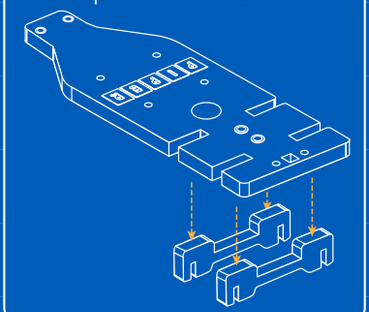


C

Detach the Motors Holder and Motors by removing Screws

6

Remove Motor Spacers from Base



DISASSEMBLY COMPLETE!

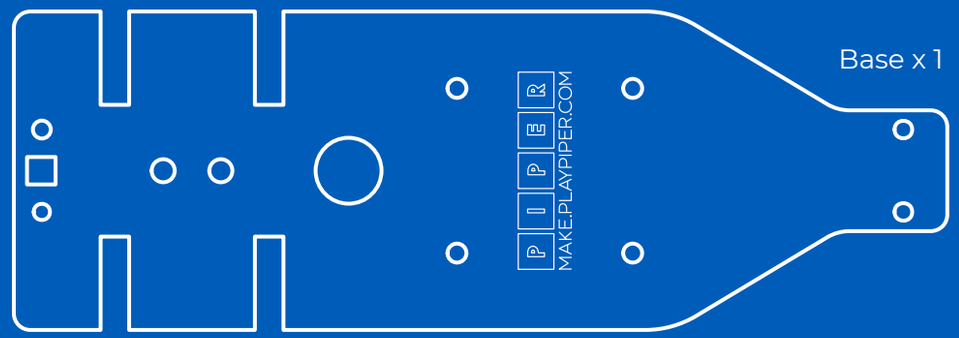
If not already done, place each piece on its spot on the Components sheet

Mission:

Piper Make
Robotic Rover

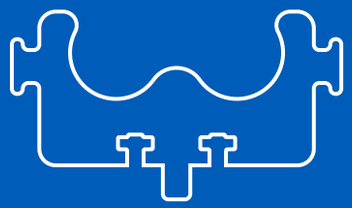
Components Sheet

Gather and organize your supplies

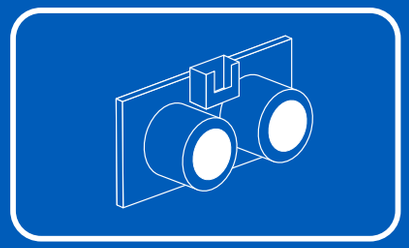


Base x 1

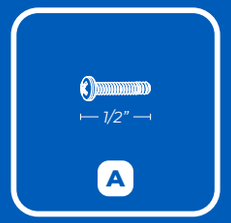
Motor Module x 1



Range Finder Holder x 1



Ultrasonic Range Finder x 1



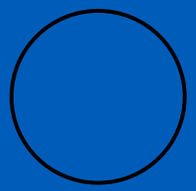
#4-40 Screw x 2

Motor Holder x 1

Spacer #1 x 2

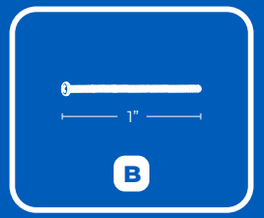
Spacer #2 x 2

Motor Spacer x 2

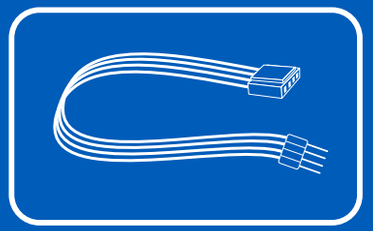


O-Ring x 1

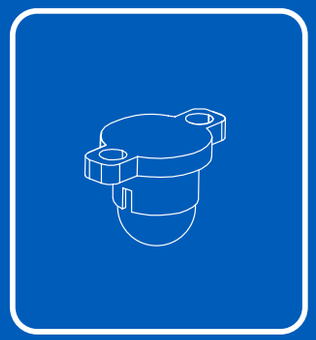
Velcro Pieces x 1



#2-56 Screw x 2

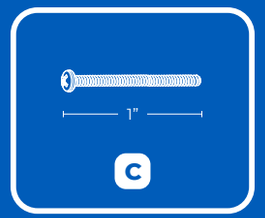


4-Pin Cable x 2



Ball and Holder x 1

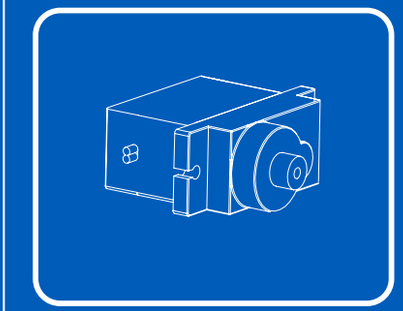
#4-40 Nut x 2



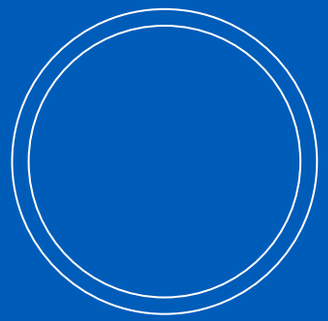
#4-40 Screw x 2

Small Screw x 2

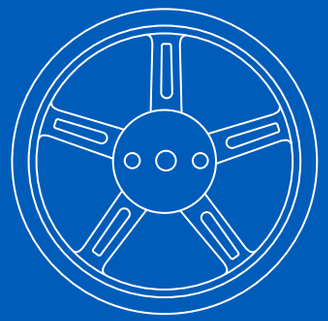
Motors and Wheels



FM90 Geared Motor x 2



Wheel x 2



Wheel Rim x 2